CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

JDL DAIRY INC

22495 W CHINA CAMP RD

LOS BANOS CA 93635

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: JDL Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
22495 W China Camp RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1979

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X160-X037-XXXX

B. OPERATORS

JDL Dairy Inc.

Operator name: JDL Dairy Inc.

Telephone no.: Landline: ; Cellular: (209) 481-4424

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2253	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

JDL Dairy Inc.

Legal owner name: JDL Dairy Inc.

Telephone no.: Landline: ; Cellular: (209) 481-4424

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2253	Los Banos	CA	93635

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	286	206	50	72
Number under roof	445	66	0	0	0	0
Maximum number	445	66	286	206	50	72
Average number	445	66	286	206	50	72
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,005.43 *tons per reporting period*

Total nitrogen from manure: 215,579.81 *lbs per reporting period*

After ammonia losses (30% loss applied): 150,905.87 *lbs per reporting period*

Total phosphorus from manure: 35,449.55 *lbs per reporting period*

Total potassium from manure: 85,806.34 *lbs per reporting period*

Total salt from manure: 224,704.95 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 41,750,810 *gallons*

Total nitrogen generated: 173,043.53 *lbs*

Total phosphorus generated: 42,020.78 *lbs*

Total potassium generated: 288,841.95 *lbs*

Total salt generated: 1,533,945.87 *lbs*

	41,750,810	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	41,750,810	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Drain Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
ARGENTIERI	22	22	2	manure	X070-X180-X001-XXXX
Brasil 1	50	50	2	process wastewater	X070-X160-X009-XXXX
Field 1	74	74	3	process wastewater	X070-X160-X029-XXXX
Field 2	67	67	3	process wastewater	X070-X160-X038-XXXX
Field 3	70	70	2	manure	X070-X160-X038-XXXX
Machado LLC	46	46	2	manure	X070-X180-X014-XXXX X070-X180-X037-XXXX
Menezes 1	70	70	2	process wastewater	X070-X170-X027-XXXX X070-X170-X029-XXXX
Menezes 2	50	50	2	process wastewater	X070-X170-X029-XXXX
Menezes 3	50	50	2	process wastewater	X070-X170-X029-XXXX X070-X170-X030-XXXX
Menezes 4	50	50	1	process wastewater	X070-X170-X029-XXXX X070-X170-X030-XXXX
Totals for areas that were used for application	549	549	21		
Totals for areas that were not used for application					
Land application area totals	549	549	21		

B. CROPS AND HARVESTS

ARGENTIERI

Field name: ARGENTIERI

09/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 09/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	267.00 *ton*	Dry-weight		73.4	14,900.00	3,400.00	25,200.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	96.20	21.95	162.70	626.28

ARGENTIERI

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	571.00 *ton*	Dry-weight		67.4	12,400.00	2,700.00	12,600.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.95	209.84	45.69	213.22	1,083.03

Brasil 1

Field name: Brasil 1

09/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 09/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	607.00 *ton*	Dry-weight		69.2	24,400.00	3,800.00	31,700.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	182.47	28.42	237.06	762.78

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,298.00 *ton*	Dry-weight		67.3	11,400.00	2,500.00	10,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.96	193.55	42.44	185.06	984.71

Field 1

Field name: Field 1

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 74 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	898.00 *ton*	Dry-weight		53.8	19,900.00	2,600.00	21,700.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	223.14	29.15	243.32	919.45

05/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 74 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/03/2023	584.00 *ton*	Dry-weight		68.6	14,300.00	0.00	200.00		5.90
09/15/2023	709.00 *ton*	Dry-weight		68.6	14,300.00	0.00	200.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	17.47	156.91	0.00	2.19	647.41

Field 2

Field name: Field 2

Field 2

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 67 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	813.00 *ton*	Dry-weight		57.4	19,600.00	2,700.00	20,300.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.13	202.63	27.91	209.87	765.05

05/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 67 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/03/2023	524.00 *ton*	Dry-weight		64.0	11,800.00	2,400.00	13,000.00		5.10
09/20/2023	642.00 *ton*	Dry-weight		64.0	11,800.00	2,400.00	13,000.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	17.40	147.86	30.07	162.89	639.04

Field 3

Field name: Field 3

Field 3

10/17/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 70 Plant date: 10/17/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/15/2023	282.00 *ton*	Dry-weight		8.7	38,000.00	2,700.00	11,400.00		7.30
09/06/2023	275.00 *ton*	Dry-weight		8.7	38,000.00	2,700.00	11,400.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.96	552.13	39.23	165.64	1,060.67

Machado LLC

Field name: Machado LLC

09/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 46 Plant date: 09/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	558.00 *ton*	Dry-weight		72.0	20,900.00	3,500.00	26,600.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.13	141.97	23.78	180.69	672.51

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 46 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,195.00 *ton*	Dry-weight		69.9	13,600.00	2,600.00	13,300.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.98	212.69	40.66	208.00	1,063.45

Menezes 1

Field name: Menezes 1

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	850.00 *ton*	Dry-weight		61.8	18,800.00	2,700.00	16,800.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	174.41	25.05	155.86	714.34

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,818.00 *ton*	Dry-weight		66.7	13,700.00	2,800.00	12,150.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.97	236.97	48.43	210.16	985.93

Menezes 2

Field name: Menezes 2

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	607.00 *ton*	Dry-weight		70.4	22,000.00	3,000.00	21,300.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	158.11	21.56	153.08	610.88

Menezes 2

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,298.00 *ton*	Dry-weight		66.7	13,700.00	2,800.00	12,150.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.96	236.86	48.41	210.07	985.49

Menezes 3

Field name: Menezes 3

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	607.00 *ton*	Dry-weight		67.0	15,700.00	3,200.00	21,300.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.14	125.79	25.64	170.66	681.05

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,298.00 *ton*	Dry-weight		66.7	13,700.00	2,800.00	12,150.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.96	236.86	48.41	210.07	985.49

Menezes 4

Field name: Menezes 4

05/28/2020: Jose Grass

Crop: Jose Grass Acres planted: 50 Plant date: 05/28/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	306.00 *ton*	Dry-weight		69.3	14,600.00	3,400.00	16,700.00		6.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	120.00	8.00	8.00	0.00
Total actual harvest content	6.12	54.86	12.78	62.75	257.78

NUTRIENT BUDGET

A. LAND APPLICATIONS

ARGENTIERI - 09/05/2022: Oats, silage-soft dough

Field name: ARGENTIERI

Crop: Oats, silage-soft dough Plant date: 09/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	119.51	22.95	54.61	0.00	431.00 *ton*
Application event totals		119.51	22.95	54.61	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.08	0.00	0.00	21.60	284,683.00 *gal*
Drain Water	Surface water	2.02	0.00	0.00	0.00	1,485,000.00 *gal*
Application event totals		2.09	0.00	0.00	21.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.31	0.00	0.00	88.79	1,170,364.00 *gal*
Drain Water	Surface water	0.45	0.00	0.00	0.00	330,000.00 *gal*
Application event totals		0.76	0.00	0.00	88.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.06	0.00	0.00	16.80	221,420.00 *gal*
Drain Water	Surface water	2.24	0.00	0.00	0.00	1,650,000.00 *gal*
Application event totals		2.30	0.00	0.00	16.80	

ARGENTIERI - 05/01/2023: Corn, silage

ARGENTIERI - 05/01/2023: Corn, silage

Field name: ARGENTIERI

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	268.88	53.20	142.18	0.00	429.00 *ton*
Application event totals		268.88	53.20	142.18	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.28	0.00	0.00	366.35	4,829,071.00 *gal*
Application event totals		1.28	0.00	0.00	366.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	5,208,649.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.67	0.00	0.00	192.77	2,541,061.00 *gal*
Application event totals		0.67	0.00	0.00	192.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	5,208,649.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.53	0.00	0.00	152.78	2,013,870.00 *gal*
Application event totals		0.53	0.00	0.00	152.78	

ARGENTIERI - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	5,208,649.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Brasil 1 - 09/05/2022: Oats, silage-soft dough

Field name: Brasil 1

Crop: Oats, silage-soft dough

Plant date: 09/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.31	26.67	146.48	787.70	828,000.00 *gal*
Canal	Surface water	0.08	0.00	0.00	21.76	651,800.00 *gal*
Drain Water	Surface water	1.52	0.00	0.00	0.00	2,550,000.00 *gal*
Application event totals		89.91	26.67	146.48	809.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.31	26.67	146.48	787.70	828,000.00 *gal*
Canal	Surface water	0.42	0.00	0.00	120.75	3,617,490.00 *gal*
Application event totals		88.73	26.67	146.48	908.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.61	16.56	102.48	587.38	936,000.00 *gal*
Canal	Surface water	0.06	0.00	0.00	17.41	521,440.00 *gal*
Drain Water	Surface water	1.77	0.00	0.00	0.00	2,958,000.00 *gal*
Application event totals		67.44	16.56	102.48	604.79	

Brasil 1 - 05/01/2023: Corn, silage

Field name: Brasil 1

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.99	13.37	82.77	474.42	756,000.00 *gal*
Canal	Surface water	0.51	0.00	0.00	145.77	4,367,060.00 *gal*
Drain Water	Surface water	4.27	0.00	0.00	0.00	7,140,000.00 *gal*
Application event totals		57.77	13.37	82.77	620.20	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.99	13.37	82.77	474.42	756,000.00 *gal*
Canal	Surface water	1.59	0.00	0.00	454.18	13,606,325.00 *gal*
Application event totals		54.58	13.37	82.77	928.60	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.26	11.91	127.38	591.23	720,000.00 *gal*
Canal	Surface water	1.59	0.00	0.00	454.18	13,606,325.00 *gal*
Application event totals		69.85	11.91	127.38	1,045.41	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	7.01	0.00	0.00	0.00	11,730,000.00 *gal*
Application event totals		7.01	0.00	0.00	0.00	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.26	11.91	127.38	591.23	720,000.00 *gal*
Drain Water	Surface water	4.88	0.00	0.00	0.00	8,160,000.00 *gal*
Application event totals		73.13	11.91	127.38	591.23	

Field 1 - 10/18/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.30	28.78	158.09	850.12	1,322,553.00 *gal*
Canal	Surface water	0.04	0.00	0.00	11.19	496,015.00 *gal*
Drain Water	Surface water	3.35	0.00	0.00	0.00	8,297,447.00 *gal*
Application event totals		98.69	28.78	158.09	861.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.44	24.84	153.75	881.23	2,078,298.00 *gal*
Application event totals		98.44	24.84	153.75	881.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.44	24.84	153.75	881.23	2,078,298.00 *gal*
Canal	Surface water	0.07	0.00	0.00	19.67	872,303.00 *gal*
Drain Water	Surface water	1.67	0.00	0.00	0.00	4,148,723.00 *gal*
Application event totals		100.18	24.84	153.75	900.91	

Field 1 - 05/25/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 05/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.74	11.29	69.88	400.56	944,681.00 *gal*
Canal	Surface water	0.34	0.00	0.00	97.98	4,344,409.00 *gal*
Drain Water	Surface water	5.40	0.00	0.00	0.00	13,382,979.00 *gal*
Application event totals		50.49	11.29	69.88	498.54	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.79	9.03	55.91	320.45	755,745.00 *gal*
Canal	Surface water	1.69	0.00	0.00	482.25	21,382,148.00 *gal*
Application event totals		37.48	9.03	55.91	802.70	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.51	10.56	112.92	524.14	944,681.00 *gal*
Canal	Surface water	2.05	0.00	0.00	584.82	25,929,621.00 *gal*
Application event totals		62.56	10.56	112.92	1,108.96	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.41	8.45	90.34	419.31	755,745.00 *gal*
Drain Water	Surface water	7.63	0.00	0.00	0.00	18,892,413.00 *gal*
Application event totals		56.04	8.45	90.34	419.31	

Field 2 - 10/15/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 10/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.30	28.78	158.09	850.12	1,197,447.00 *gal*
Canal	Surface water	0.04	0.00	0.00	11.19	449,085.00 *gal*
Drain Water	Surface water	3.35	0.00	0.00	0.00	7,512,553.00 *gal*
Application event totals		98.69	28.78	158.09	861.31	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.54	20.33	125.79	721.01	1,539,574.00 *gal*
Application event totals		80.54	20.33	125.79	721.01	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.44	24.84	153.75	881.23	1,881,702.00 *gal*
Canal	Surface water	0.07	0.00	0.00	19.67	789,787.00 *gal*
Drain Water	Surface water	1.67	0.00	0.00	0.00	3,756,277.00 *gal*
Application event totals		100.18	24.84	153.75	900.91	

Field 2 - 05/25/2023: Sudangrass, silage

Field name: Field 2

Crop: Sudangrass, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.74	11.29	69.88	400.56	855,319.00 *gal*
Canal	Surface water	0.34	0.00	0.00	97.98	3,933,451.00 *gal*
Drain Water	Surface water	5.40	0.00	0.00	0.00	12,117,021.00 *gal*
Application event totals		50.49	11.29	69.88	498.54	

Field 2 - 05/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.79	9.03	55.91	320.45	684,255.00 *gal*
Canal	Surface water	1.72	0.00	0.00	490.31	19,682,742.00 *gal*
Application event totals		37.51	9.03	55.91	810.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.41	8.45	90.34	419.31	684,255.00 *gal*
Canal	Surface water	2.05	0.00	0.00	584.82	23,476,819.00 *gal*
Application event totals		50.45	8.45	90.34	1,004.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.41	8.45	90.34	419.31	684,255.00 *gal*
Drain Water	Surface water	7.63	0.00	0.00	0.00	17,104,387.00 *gal*
Application event totals		56.03	8.45	90.34	419.31	

Field 3 - 10/17/2019: Alfalfa, hay

Field name: Field 3

Crop: Alfalfa, hay

Plant date: 10/17/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.32	0.00	0.00	90.14	3,780,440.00 *gal*
Application event totals		0.32	0.00	0.00	90.14	

Field 3 - 10/17/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	3.26	0.00	0.00	0.00	7,650,000.00 *gal*
Application event totals		3.26	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	18.65	782,160.00 *gal*
Drain Water	Surface water	2.18	0.00	0.00	0.00	5,100,000.00 *gal*
Application event totals		2.24	0.00	0.00	18.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.61	0.00	0.00	174.83	7,332,750.00 *gal*
Application event totals		0.61	0.00	0.00	174.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.05	0.00	0.00	584.33	24,507,680.00 *gal*
Application event totals		2.05	0.00	0.00	584.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	3.26	0.00	0.00	0.00	7,650,000.00 *gal*
Application event totals		3.26	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	3.26	0.00	0.00	0.00	7,650,000.00 *gal*
Application event totals		3.26	0.00	0.00	0.00	

Field 3 - 10/17/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.05	0.00	0.00	584.33	24,507,680.00 *gal*
Application event totals		2.05	0.00	0.00	584.33	

11/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	358.51	70.93	189.57	0.00	1,820.00 *ton*
Application event totals		358.51	70.93	189.57	0.00	

Machado LLC - 09/05/2022: Oats, silage-soft dough

Field name: Machado LLC

Crop: Oats, silage-soft dough

Plant date: 09/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	158.61	30.46	72.48	0.00	1,196.00 *ton*
Application event totals		158.61	30.46	72.48	0.00	

09/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.08	0.00	0.00	21.60	595,247.00 *gal*
Drain Water	Surface water	2.02	0.00	0.00	0.00	3,105,000.00 *gal*
Application event totals		2.09	0.00	0.00	21.60	

Machado LLC - 09/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.31	0.00	0.00	88.79	2,447,126.00 *gal*
Drain Water	Surface water	0.45	0.00	0.00	0.00	690,000.00 *gal*
Application event totals		0.76	0.00	0.00	88.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.06	0.00	0.00	16.80	462,970.00 *gal*
Drain Water	Surface water	2.24	0.00	0.00	0.00	3,450,000.00 *gal*
Application event totals		2.30	0.00	0.00	16.80	

Machado LLC - 05/01/2023: Corn, silage

Field name: Machado LLC

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	284.77	56.34	150.58	0.00	950.00 *ton*
Application event totals		284.77	56.34	150.58	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.28	0.00	0.00	366.35	10,097,149.00 *gal*
Application event totals		1.28	0.00	0.00	366.35	

Machado LLC - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	10,890,811.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.67	0.00	0.00	192.77	5,313,129.00 *gal*
Application event totals		0.67	0.00	0.00	192.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	10,890,811.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.53	0.00	0.00	152.78	4,210,820.00 *gal*
Application event totals		0.53	0.00	0.00	152.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.38	0.00	0.00	395.15	10,890,811.00 *gal*
Application event totals		1.38	0.00	0.00	395.15	

Menezes 1 - 10/18/2022: Oats, silage-soft dough

Field name: Menezes 1

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Menezes 1 - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.98	28.99	159.22	856.20	1,260,000.00 *gal*
Canal	Surface water	0.31	0.00	0.00	89.75	3,764,145.00 *gal*
Application event totals		96.30	28.99	159.22	945.94	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.82	17.37	107.50	616.13	1,374,545.00 *gal*
Application event totals		68.82	17.37	107.50	616.13	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.82	17.37	107.50	616.13	1,374,545.00 *gal*
Canal	Surface water	0.21	0.00	0.00	61.32	2,571,647.00 *gal*
Application event totals		69.04	17.37	107.50	677.45	

Menezes 1 - 05/25/2023: Corn, silage

Field name: Menezes 1

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	1,145,455.00 *gal*
Canal	Surface water	0.93	0.00	0.00	265.24	11,124,692.00 *gal*
Application event totals		58.28	14.47	89.58	778.69	

Menezes 1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	1,145,455.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	14,667,417.00 *gal*
Application event totals		58.58	14.47	89.58	863.16	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	1,145,455.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	14,667,417.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	14,667,417.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	14,667,417.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	1,145,455.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	14,667,417.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

Menezes 2 - 10/20/2022: Oats, silage-soft dough

Field name: Menezes 2

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Menezes 2 - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.71	31.63	173.70	934.03	981,818.00 *gal*
Canal	Surface water	0.31	0.00	0.00	89.75	2,688,675.00 *gal*
Application event totals		105.02	31.63	173.70	1,023.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.76	23.16	143.33	821.51	1,309,091.00 *gal*
Canal	Surface water	0.21	0.00	0.00	61.32	1,836,891.00 *gal*
Application event totals		91.98	23.16	143.33	882.83	

Menezes 2 - 05/25/2023: Corn, silage

Field name: Menezes 2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	818,182.00 *gal*
Canal	Surface water	0.93	0.00	0.00	265.24	7,946,209.00 *gal*
Application event totals		58.28	14.47	89.58	778.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		58.58	14.47	89.58	863.16	

Menezes 2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

Menezes 3 - 10/18/2022: Oats, silage-soft dough

Field name: Menezes 3

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Menezes 3 - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.98	28.99	159.22	856.20	900,000.00 *gal*
Canal	Surface water	0.31	0.00	0.00	89.75	2,688,675.00 *gal*
Application event totals		96.30	28.99	159.22	945.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.82	17.37	107.50	616.13	981,818.00 *gal*
Canal	Surface water	0.21	0.00	0.00	61.32	1,836,891.00 *gal*
Application event totals		69.04	17.37	107.50	677.45	

Menezes 3 - 05/25/2023: Corn, silage

Field name: Menezes 3

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	818,182.00 *gal*
Canal	Surface water	0.93	0.00	0.00	265.24	7,946,209.00 *gal*
Application event totals		58.28	14.47	89.58	778.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.35	14.47	89.58	513.45	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		58.58	14.47	89.58	863.16	

Menezes 3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.56	13.53	144.75	671.85	818,182.00 *gal*
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		78.79	13.53	144.75	1,021.56	

Menezes 4 - 05/28/2020: Jose Grass

Field name: Menezes 4

Crop: Jose Grass

Plant date: 05/28/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Menezes 4 - 05/28/2020: Jose Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.36	15.81	86.85	467.02	490,909.00 *gal*
Canal	Surface water	0.31	0.00	0.00	89.75	2,688,675.00 *gal*
Application event totals		52.67	15.81	86.85	556.76	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.21	0.00	0.00	61.32	1,836,891.00 *gal*
Application event totals		0.21	0.00	0.00	61.32	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	265.24	7,946,209.00 *gal*
Application event totals		0.93	0.00	0.00	265.24	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

Menezes 4 - 05/28/2020: Jose Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

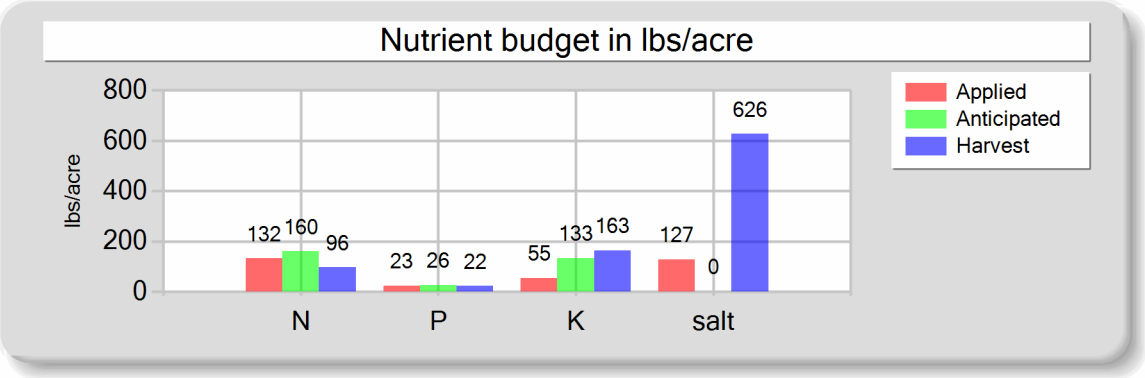
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	349.71	10,476,726.00 *gal*
Application event totals		1.22	0.00	0.00	349.71	

B. NUTRIENT BUDGET

ARGENTIERI - 09/05/2022: Oats, silage-soft dough

Field name: ARGENTIERI Crop: Oats, silage-soft dough Plant date: 09/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	119.51	22.95	54.61	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.15	0.00	0.00	127.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	131.66	22.95	54.61	127.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	96.20	21.95	162.70	626.28
Nutrient balance	35.46	1.00	-108.09	-499.10
Applied to removed ratio	1.37	1.05	0.34	0.20

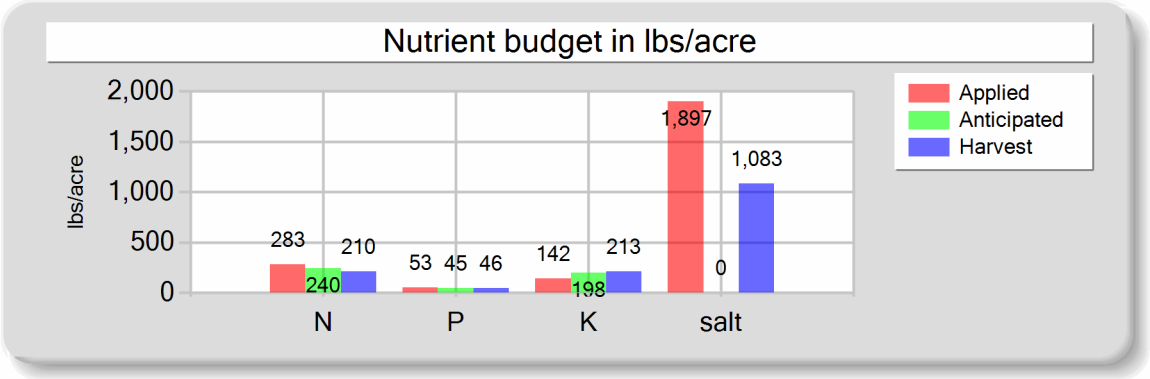
Fresh water applied	
5,141,467.00	*gallons*
189.34	*acre-inches*
8.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

ARGENTIERI - 05/01/2023: Corn, silage

Field name: ARGENTIERI Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	268.88	53.20	142.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.64	0.00	0.00	1,897.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.52	53.20	142.18	1,897.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.84	45.69	213.22	1,083.03
Nutrient balance	72.69	7.51	-71.04	814.31
Applied to removed ratio	1.35	1.16	0.67	1.75

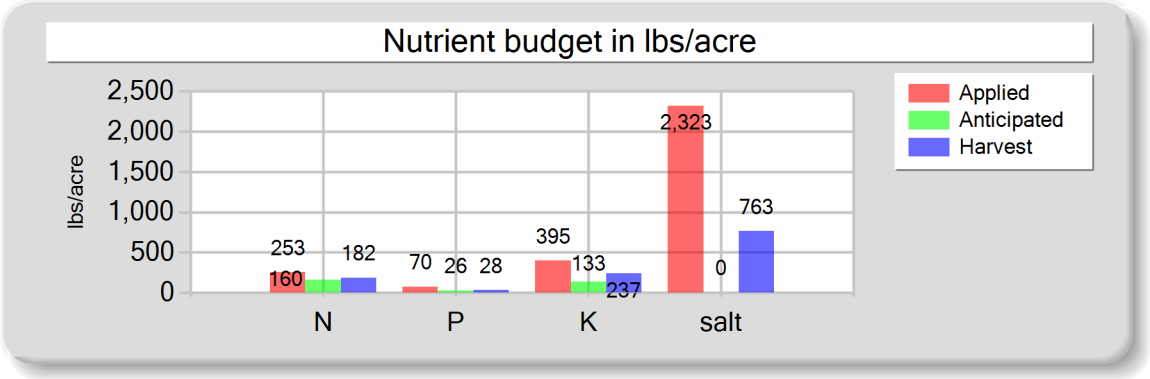
Fresh water applied	
25,009,949.00	*gallons*
921.03	*acre-inches*
41.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Brasil 1 - 09/05/2022: Oats, silage-soft dough

Field name: Brasil 1 Crop: Oats, silage-soft dough Plant date: 09/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.22	69.90	395.45	2,162.78
Fresh water	3.85	0.00	0.00	159.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.07	69.90	395.45	2,322.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	182.47	28.42	237.06	762.78
Nutrient balance	70.60	41.48	158.39	1,559.92
Applied to removed ratio	1.39	2.46	1.67	3.05

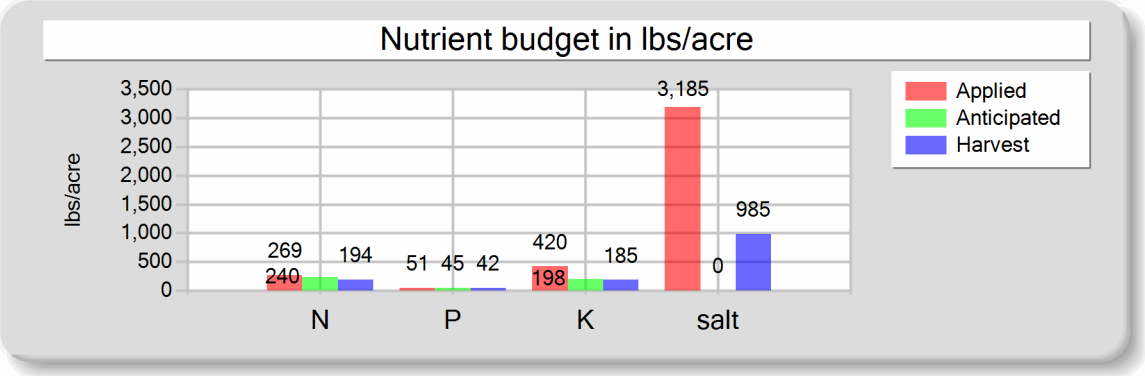
Fresh water applied	
10,298,730.00	*gallons*
379.27	*acre-inches*
7.59	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Brasil 1 - 05/01/2023: Corn, silage

Field name: Brasil 1 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.50	50.57	420.30	2,131.30
Fresh water	19.84	0.00	0.00	1,054.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.34	50.57	420.30	3,185.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	193.55	42.44	185.06	984.71
Nutrient balance	75.79	8.12	235.24	2,200.72
Applied to removed ratio	1.39	1.19	2.27	3.23

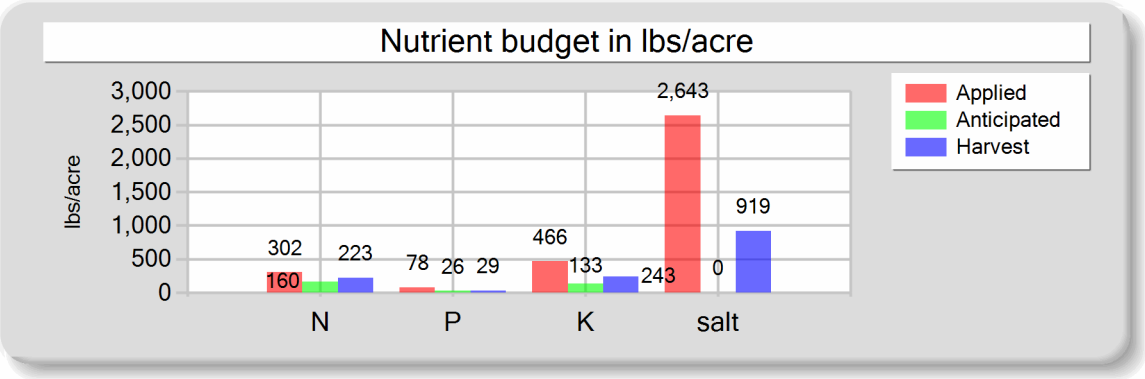
Fresh water applied	
58,609,710.00	*gallons*
2,158.40	*acre-inches*
43.17	*inches/acre*

Process wastewater applied	
2,952,000.00	*gallons*
108.71	*acre-inches*
2.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 10/18/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.17	78.47	465.59	2,612.59
Fresh water	5.13	0.00	0.00	30.86
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	301.97	78.47	465.59	2,643.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	223.14	29.15	243.32	919.45
Nutrient balance	78.84	49.32	222.27	1,723.99
Applied to removed ratio	1.35	2.69	1.91	2.88

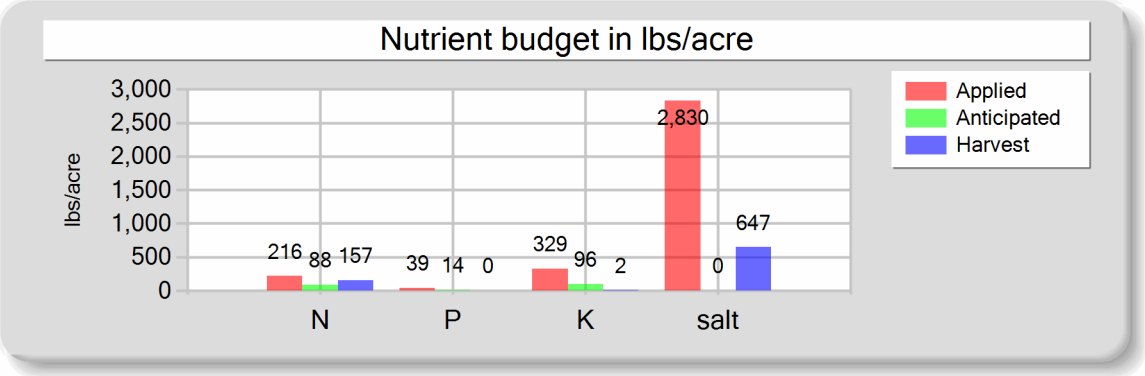
Fresh water applied	
13,814,488.00	*gallons*
508.74	*acre-inches*
6.87	*inches/acre*

Process wastewater applied	
5,479,149.00	*gallons*
201.78	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/25/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.46	39.33	329.06	1,664.46
Fresh water	17.11	0.00	0.00	1,165.06
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	215.90	39.33	329.06	2,829.51
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	156.91	0.00	2.19	647.41
Nutrient balance	58.98	39.33	326.86	2,182.10
Applied to removed ratio	1.38	0.00	149.94	4.37

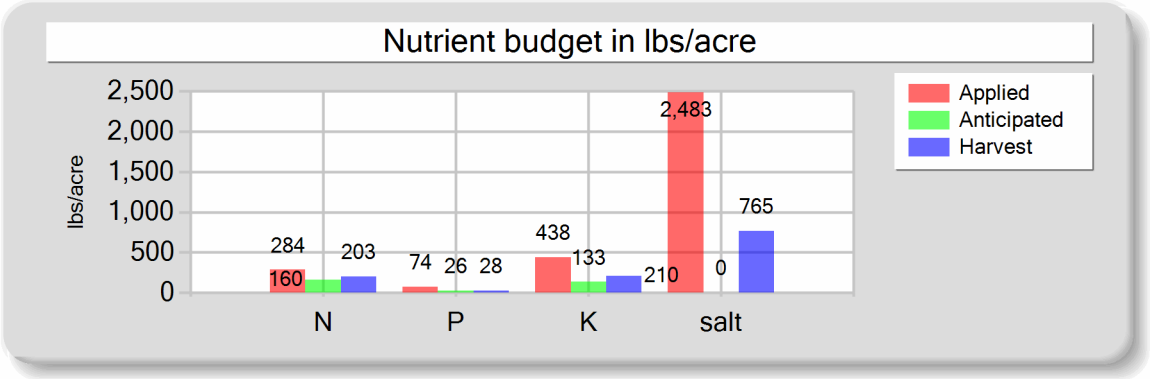
Fresh water applied	
83,931,570.00	*gallons*
3,090.91	*acre-inches*
41.77	*inches/acre*

Process wastewater applied	
3,400,852.00	*gallons*
125.24	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field 2 - 10/15/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.28	73.95	437.63	2,452.36
Fresh water	5.13	0.00	0.00	30.86
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	284.08	73.95	437.63	2,483.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	202.63	27.91	209.87	765.05
Nutrient balance	81.44	46.04	227.76	1,718.18
Applied to removed ratio	1.40	2.65	2.09	3.25

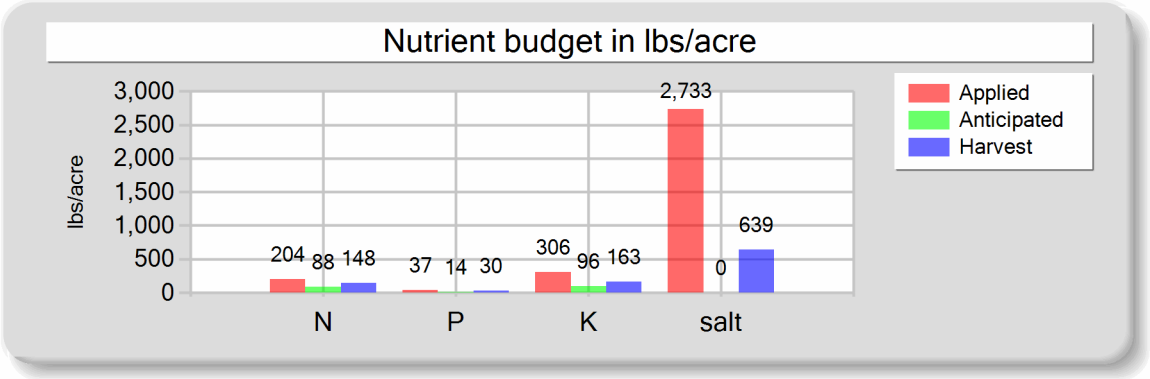
Fresh water applied	
12,507,702.00	*gallons*
460.62	*acre-inches*
6.87	*inches/acre*

Process wastewater applied	
4,618,723.00	*gallons*
170.09	*acre-inches*
2.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/25/2023: Sudangrass, silage

Field name: Field 2 Crop: Sudangrass, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	177.35	37.22	306.47	1,559.63
Fresh water	17.14	0.00	0.00	1,173.11
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	203.82	37.22	306.47	2,732.73
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	147.86	30.07	162.89	639.04
Nutrient balance	55.97	7.15	143.58	2,093.70
Applied to removed ratio	1.38	1.24	1.88	4.28

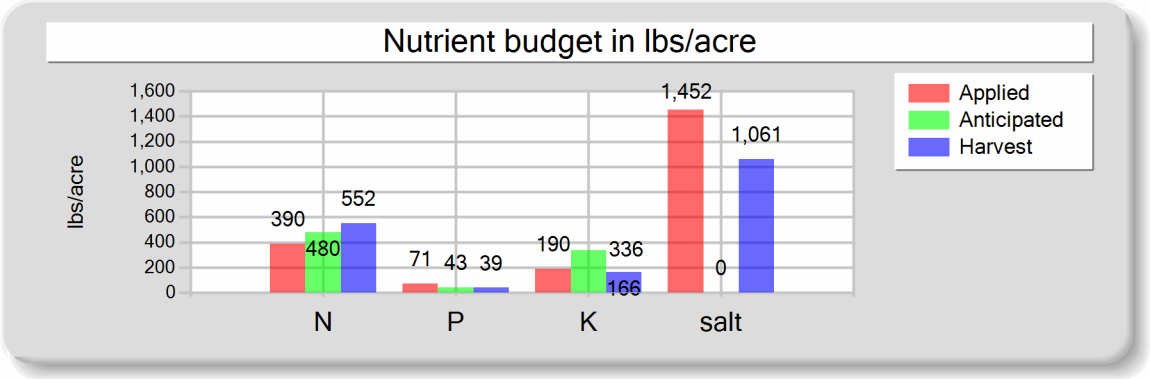
Fresh water applied	
76,314,420.00	*gallons*
2,810.40	*acre-inches*
41.95	*inches/acre*

Process wastewater applied	
2,908,084.00	*gallons*
107.09	*acre-inches*
1.60	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field 3 - 10/17/2019: Alfalfa, hay

Field name: Field 3 Crop: Alfalfa, hay Plant date: 10/17/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	358.51	70.93	189.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	17.05	0.00	0.00	1,452.29
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	389.56	70.93	189.57	1,452.29
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	552.13	39.23	165.64	1,060.67
Nutrient balance	-162.57	31.70	23.93	391.61
Applied to removed ratio	0.71	1.81	1.14	1.37

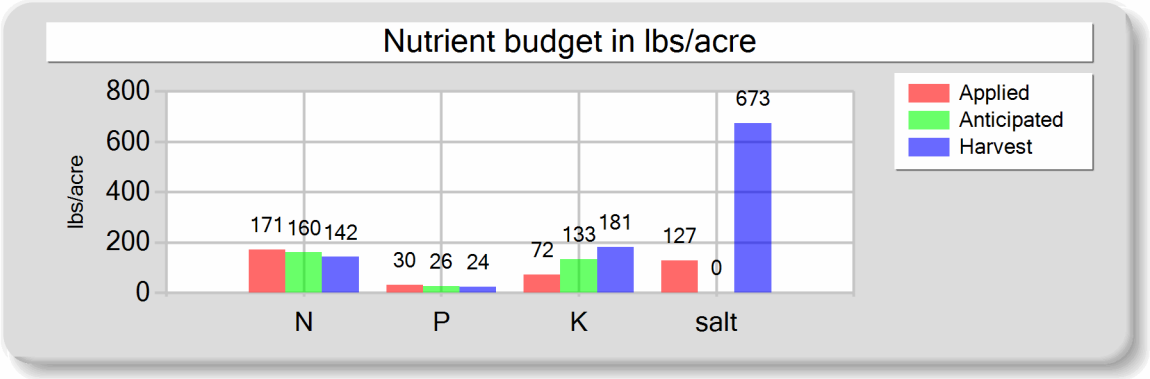
Fresh water applied	
88,960,710.00	*gallons*
3,276.12	*acre-inches*
46.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

Machado LLC - 09/05/2022: Oats, silage-soft dough

Field name: Machado LLC Crop: Oats, silage-soft dough Plant date: 09/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.61	30.46	72.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.15	0.00	0.00	127.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.76	30.46	72.48	127.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.97	23.78	180.69	672.51
Nutrient balance	28.79	6.69	-108.22	-545.33
Applied to removed ratio	1.20	1.28	0.40	0.19

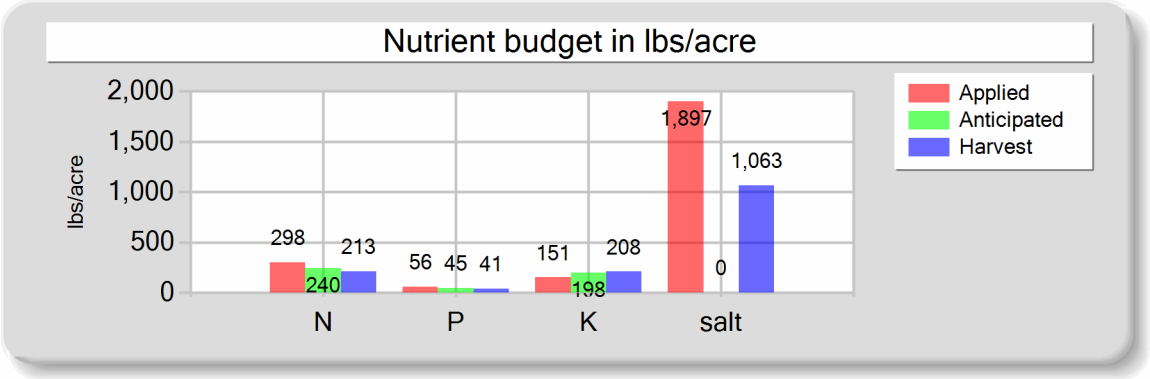
Fresh water applied	
10,750,343.00	*gallons*
395.90	*acre-inches*
8.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Machado LLC - 05/01/2023: Corn, silage

Field name: Machado LLC Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.77	56.34	150.58	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.64	0.00	0.00	1,897.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.41	56.34	150.58	1,897.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.69	40.66	208.00	1,063.45
Nutrient balance	85.72	15.68	-57.42	833.90
Applied to removed ratio	1.40	1.39	0.72	1.78

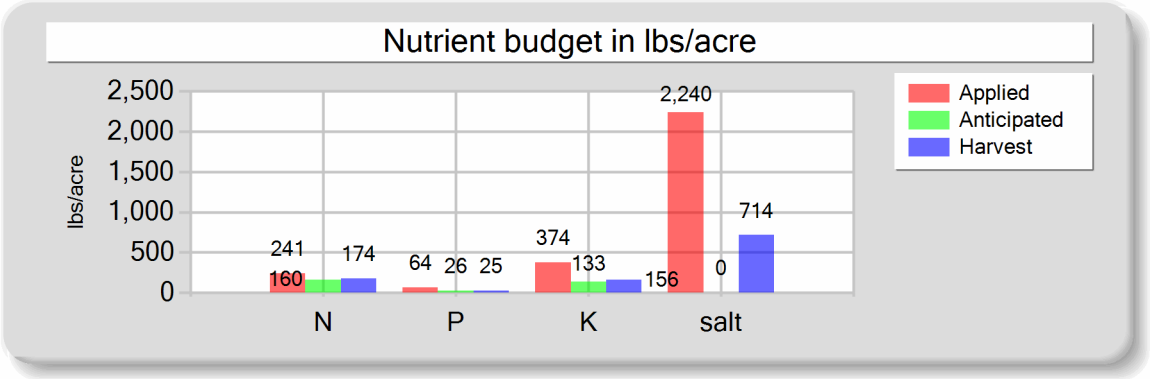
Fresh water applied	
52,293,531.00	*gallons*
1,925.79	*acre-inches*
41.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 1 - 10/18/2022: Oats, silage-soft dough

Field name: Menezes 1 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	233.63	63.73	374.21	2,088.46
Fresh water	0.53	0.00	0.00	151.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.16	63.73	374.21	2,239.53
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	174.41	25.05	155.86	714.34
Nutrient balance	66.75	38.68	218.36	1,525.19
Applied to removed ratio	1.38	2.54	2.40	3.14

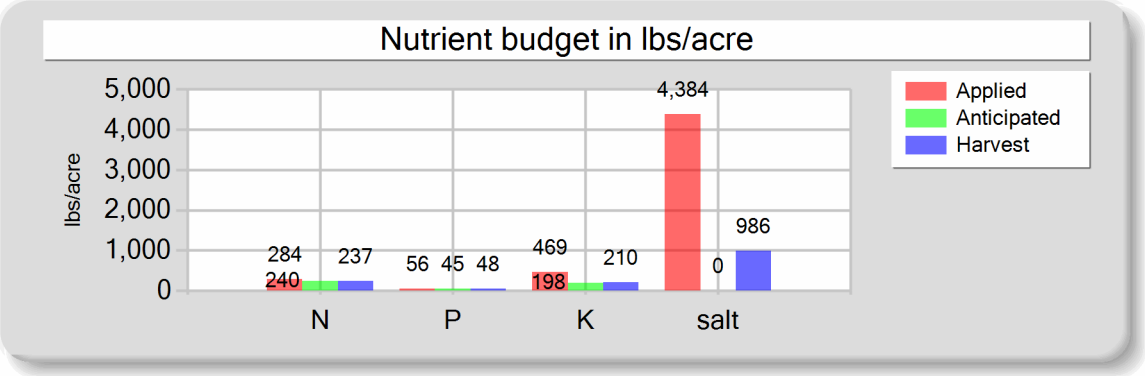
Fresh water applied	
6,335,792.00	*gallons*
233.33	*acre-inches*
3.33	*inches/acre*

Process wastewater applied	
4,009,090.00	*gallons*
147.64	*acre-inches*
2.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 1 - 05/25/2023: Corn, silage

Field name: Menezes 1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.83	56.01	468.66	2,370.59
Fresh water	7.05	0.00	0.00	2,013.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.88	56.01	468.66	4,384.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.97	48.43	210.16	985.93
Nutrient balance	46.91	7.58	258.50	3,398.47
Applied to removed ratio	1.20	1.16	2.23	4.45

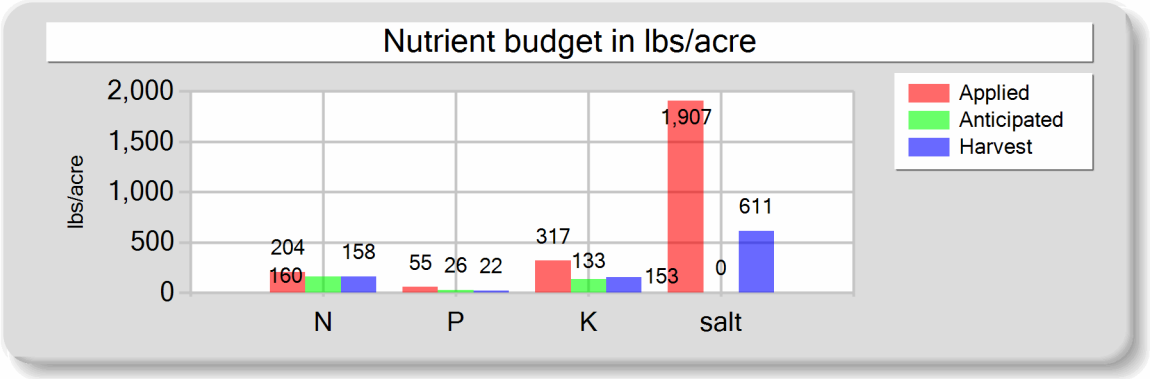
Fresh water applied	
84,461,777.00	*gallons*
3,110.44	*acre-inches*
44.43	*inches/acre*

Process wastewater applied	
4,581,820.00	*gallons*
168.73	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 2 - 10/20/2022: Oats, silage-soft dough

Field name: Menezes 2 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.47	54.79	317.03	1,755.55
Fresh water	0.53	0.00	0.00	151.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	204.00	54.79	317.03	1,906.61
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.11	21.56	153.08	610.88
Nutrient balance	45.89	33.23	163.94	1,295.72
Applied to removed ratio	1.29	2.54	2.07	3.12

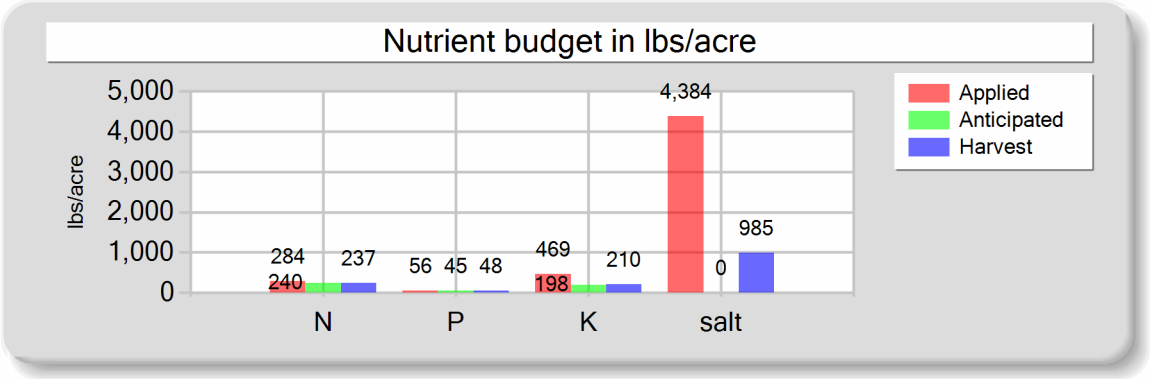
Fresh water applied	
4,525,566.00	*gallons*
166.66	*acre-inches*
3.33	*inches/acre*

Process wastewater applied	
2,290,909.00	*gallons*
84.37	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 2 - 05/25/2023: Corn, silage

Field name: Menezes 2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.83	56.01	468.66	2,370.59
Fresh water	7.05	0.00	0.00	2,013.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.88	56.01	468.66	4,384.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.86	48.41	210.07	985.49
Nutrient balance	47.02	7.60	258.59	3,398.90
Applied to removed ratio	1.20	1.16	2.23	4.45

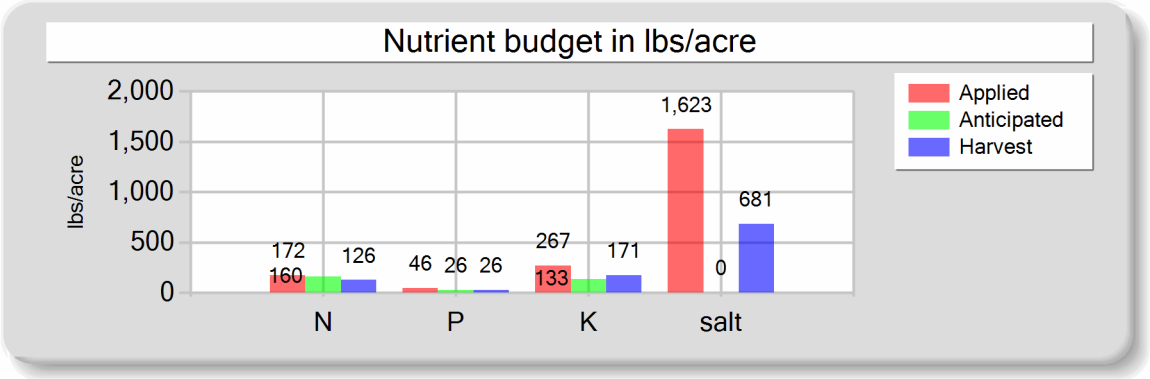
Fresh water applied	
60,329,839.00	*gallons*
2,221.74	*acre-inches*
44.43	*inches/acre*

Process wastewater applied	
3,272,728.00	*gallons*
120.52	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 3 - 10/18/2022: Oats, silage-soft dough

Field name: Menezes 3 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.81	46.36	266.72	1,472.33
Fresh water	0.53	0.00	0.00	151.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.34	46.36	266.72	1,623.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.79	25.64	170.66	681.05
Nutrient balance	46.54	20.72	96.05	942.34
Applied to removed ratio	1.37	1.81	1.56	2.38

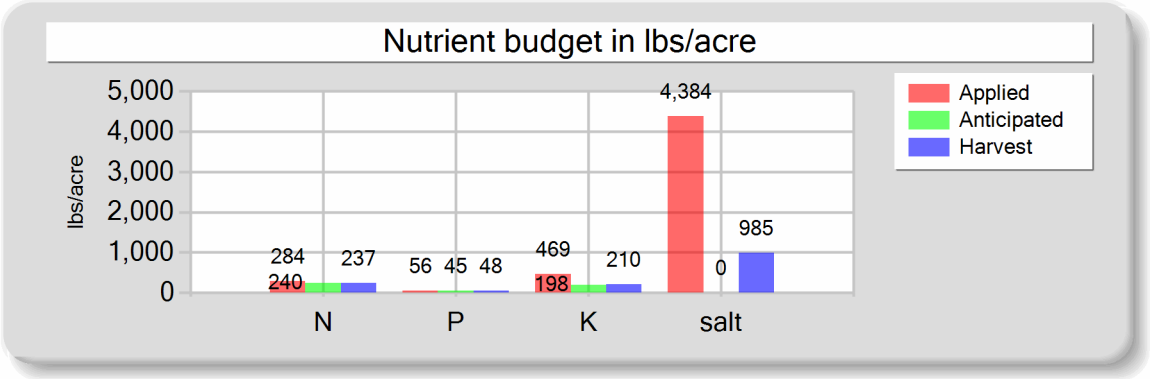
Fresh water applied	
4,525,566.00	*gallons*
166.66	*acre-inches*
3.33	*inches/acre*

Process wastewater applied	
1,881,818.00	*gallons*
69.30	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 3 - 05/25/2023: Corn, silage

Field name: Menezes 3 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.83	56.01	468.66	2,370.59
Fresh water	7.05	0.00	0.00	2,013.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.88	56.01	468.66	4,384.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.86	48.41	210.07	985.49
Nutrient balance	47.02	7.60	258.59	3,398.90
Applied to removed ratio	1.20	1.16	2.23	4.45

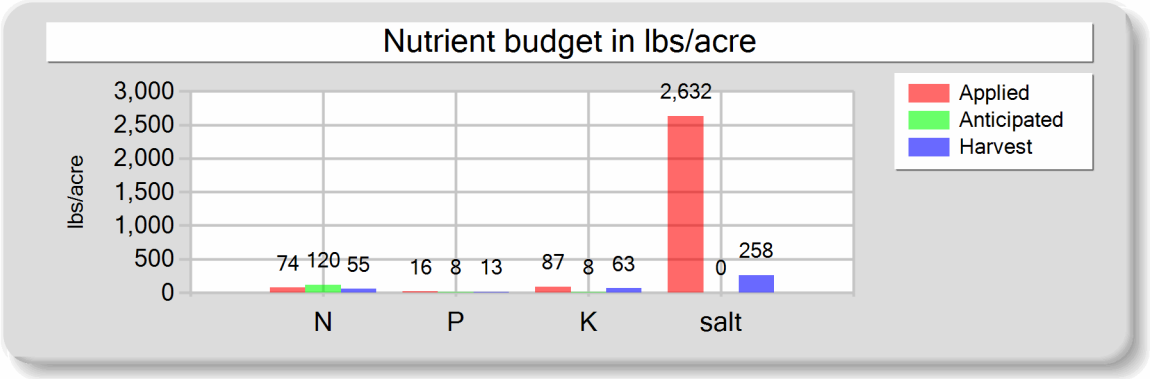
Fresh water applied	
60,329,839.00	*gallons*
2,221.74	*acre-inches*
44.43	*inches/acre*

Process wastewater applied	
3,272,728.00	*gallons*
120.52	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Menezes 4 - 05/28/2020: Jose Grass

Field name: Menezes 4 Crop: Jose Grass Plant date: 05/28/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	52.36	15.81	86.85	467.02
Fresh water	7.58	0.00	0.00	2,164.87
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	73.93	15.81	86.85	2,631.89
Anticipated crop nutrient removal	120.00	8.00	8.00	0.00
Actual crop nutrient removal	54.86	12.78	62.75	257.78
Nutrient balance	19.07	3.04	24.10	2,374.11
Applied to removed ratio	1.35	1.24	1.38	10.21

Fresh water applied	
64,855,405.00	*gallons*
2,388.40	*acre-inches*
47.77	*inches/acre*

Process wastewater applied	
490,909.00	*gallons*
18.08	*acre-inches*
0.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separator Manure

Sample and source description: 2022 Fall Separator Manure

Sample date: 10/05/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,900.00	6,900.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,200.00	8,800.00	22,600.00	21,500.00	11,500.00	8,900.00	6,800.00	11,000.00		34.70
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	5,500.00	14,700.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	637.00	471.00	0.00	2.00	193.00	1,060.00								11,600.00	5,700
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	977.00	448.00	0.00	1.30	290.00	1,030.00								10,400.00	5,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	419.00	280.00	0.00	1.00	106.00	656.00								7,890.00	3,760
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	567.00	396.00	0.00	1.00	99.10	1,060.00								10,200.00	4,920
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	465.00	297.00	0.00	0.00	93.40	775.00								7,920.00	4,850
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW-House

Sample description: DW-House

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								773.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW-MB

Sample description: DW-MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.10	0.00	1.10								1,420.00	0
DL	0.01	0.01	0.01								10.00	1

Drain Water

Drain Water

Sample description: Drain Water

Sample date: 06/08/2021 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.58	0.00	3.58								834.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

ARGENTIERI - 09/05/2022: Oats, silage-soft dough

ARGENTIERI - 09/05/2022: Oats, silage-soft dough

Argentin Oats

Sample and source description: Argentin Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,400.00	25,200.00		9.70
DL	0.01	0.01	0.01		1.00

ARGENTIERI - 05/01/2023: Corn, silage

Argentieri Corn

Sample and source description: Argentieri Corn

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,700.00	12,600.00		6.40
DL	0.01	0.01	0.01		1.00

Brasil 1 - 09/05/2022: Oats, silage-soft dough

Brasil 1 Oats

Sample and source description: Brasil 1 Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	3,800.00	31,700.00		10.20
DL	0.01	0.01	0.01		1.00

Brasil 1 - 05/01/2023: Corn, silage

Brasil 1 Corn

Sample and source description: Brasil 1 Corn

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,500.00	10,900.00		5.80
DL	0.01	0.01	0.01		1.00

Field 1 - 10/18/2022: Oats, silage-soft dough

Field 1 Oat Silage

Sample and source description: Field 1 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	2,600.00	21,700.00		8.20
DL	0.01	0.01	0.01		1.00

Field 1 - 05/25/2023: Sudangrass, silage

Field 1 Sudan Silage

Sample and source description: Field 1 Sudan Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	0.00	200.00		5.90
DL	0.01	0.01	0.01		1.00

Field 2 - 10/15/2022: Oats, silage-soft dough

Field 2 Oat Silage

Sample and source description: Field 2 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,600.00	2,700.00	20,300.00		7.40
DL	0.01	0.01	0.01		1.00

Field 2 - 05/25/2023: Sudangrass, silage

Field 2 Sudan Silage

Sample and source description: Field 2 Sudan Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,400.00	13,000.00		5.10
DL	0.01	0.01	0.01		1.00

Field 3 - 10/17/2019: Alfalfa, hay

Field 3 Alfalfa

Sample and source description: Field 3 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,000.00	2,700.00	11,400.00		7.30
DL	0.01	0.01	0.01		1.00

Machado LLC - 09/05/2022: Oats, silage-soft dough

Machado LLC Oats

Sample and source description: Machado LLC Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	3,500.00	26,600.00		9.90
DL	0.01	0.01	0.01		1.00

Machado LLC - 05/01/2023: Corn, silage

Machado LLC Corn

Sample and source description: Machado LLC Corn

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,600.00	13,300.00		6.80
DL	0.01	0.01	0.01		1.00

Menezes 1 - 10/18/2022: Oats, silage-soft dough

Meneze Ranch Field 3 Oat Silage

Sample and source description: Meneze Ranch Field 3 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	2,700.00	16,800.00		7.70
DL	0.01	0.01	0.01		1.00

Menezes 1 - 05/25/2023: Corn, silage

Menezes 1 Corn

Sample and source description: Menezes 1 Corn

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,800.00	12,150.00		5.70
DL	0.01	0.01	0.01		1.00

Menezes 2 - 10/20/2022: Oats, silage-soft dough

Meneze Ranch Field 2 Oat Silage

Sample and source description: Meneze Ranch Field 2 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	3,000.00	21,300.00		8.50
DL	0.01	0.01	0.01		1.00

Menezes 2 - 05/25/2023: Corn, silage

Menezes 2 Corn

Sample and source description: Menezes 2 Corn

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,800.00	12,150.00		5.70
DL	0.01	0.01	0.01		1.00

Menezes 3 - 10/18/2022: Oats, silage-soft dough

Meneze 3 Oats

Sample and source description: Meneze 3 Oats

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	3,200.00	21,300.00		8.50
DL	0.01	0.01	0.01		1.00

Menezes 3 - 05/25/2023: Corn, silage

Menezes 3 Corn

Sample and source description: Menezes 3 Corn

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,800.00	12,150.00		5.70
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

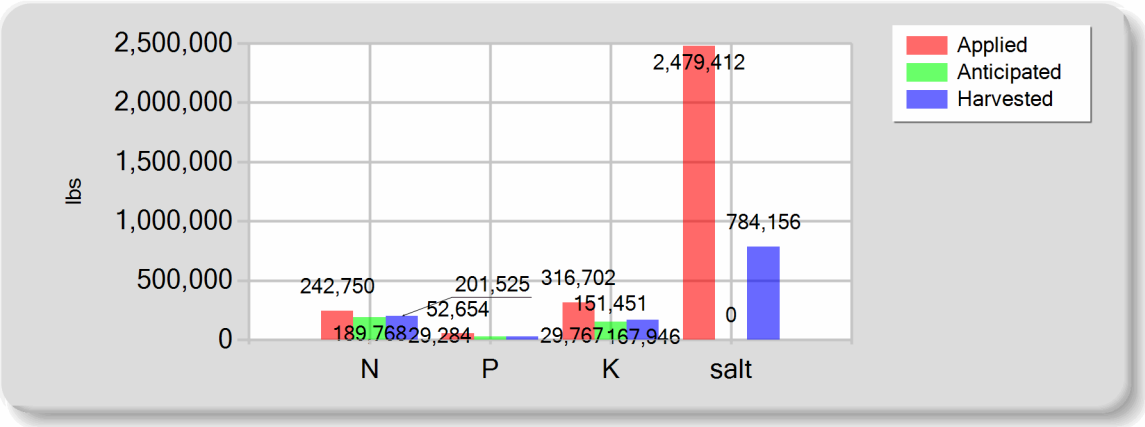
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

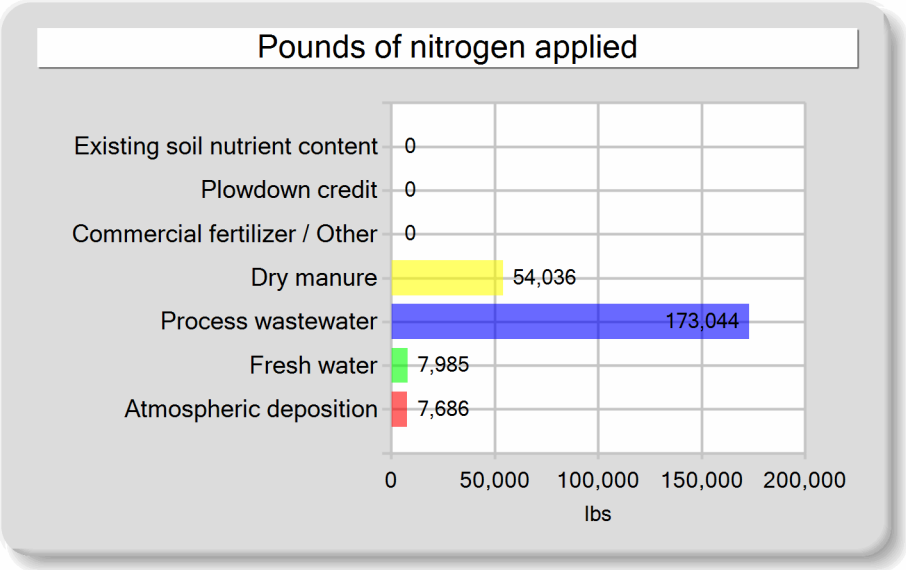
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

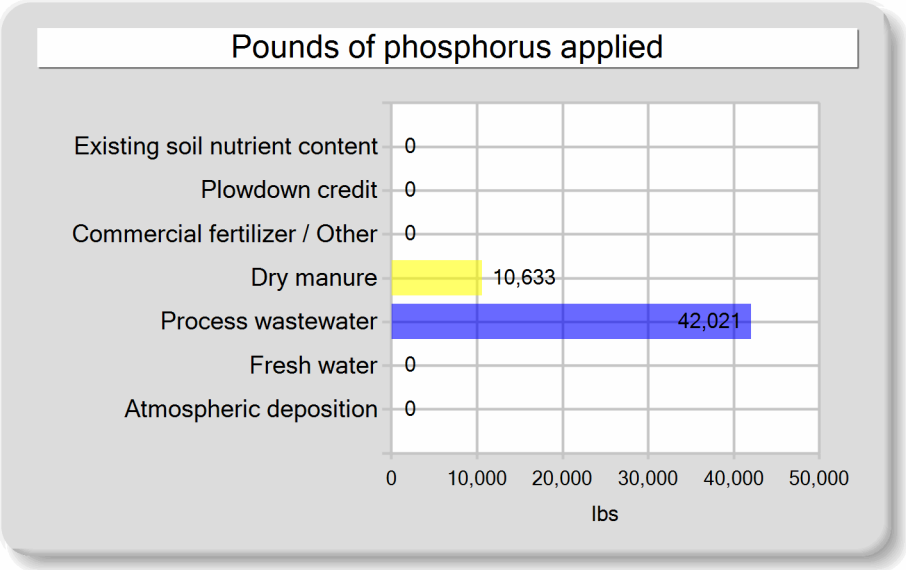
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	54,035.72	10,633.07	27,859.97	0.00
Process wastewater	173,043.53	42,020.78	288,841.95	1,533,945.87
Fresh water	7,984.88	0.00	0.00	945,466.14
Atmospheric deposition	7,686.00	0.00	0.00	0.00
Total nutrients applied	242,750.12	52,653.85	316,701.93	2,479,412.01
Anticipated crop nutrient removal	189,768.00	29,284.00	151,451.20	0.00
Actual crop nutrient removal	201,525.20	29,767.31	167,946.48	784,156.04
Nutrient balance	41,224.92	22,886.54	148,755.45	1,695,255.98
Applied to removed ratio	1.20	1.77	1.89	3.16

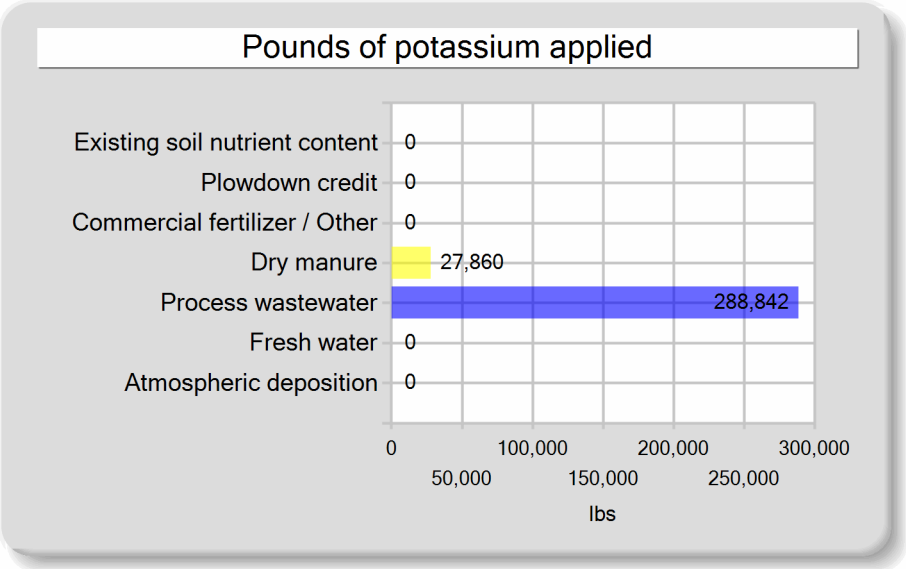
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

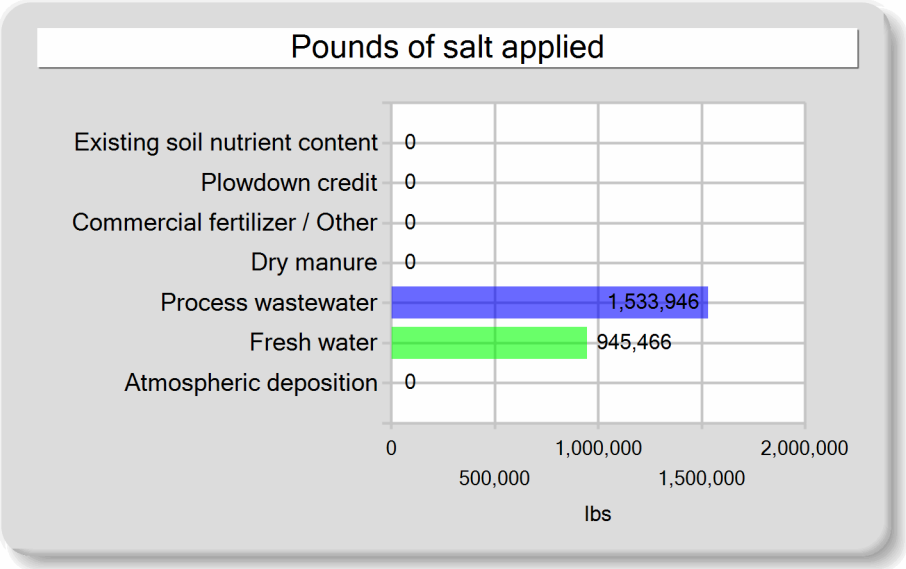


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Machado

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
JDL Dairy Inc.	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Lab No.: 23L1000
Sampled By:
Requested By: Lisa Machado
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

JDL Dairy, Inc
PO Box 2253
Los Banos CA 93635
00-0025117 08

E-mail: machdairy@yahoo.com
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	14:15	1420	1.42	1.1	ND		8.0
2	DW - House	12/13/23	14:15	773	0.77	0.2	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

JDL Dairy, Inc
PO Box 2253
Los Banos CA 93635
00-0025117 08

E-mail: machdairy@yahoo.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L1000
Sampled By:
Requested By: Lisa Machado
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	14:15	**1420**	**1.42**	1.1	3.00	0.25
2	DW - House	12/13/23	14:15	773	0.77	0.2	0.55	0.05

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

JDL Dairy, Inc
PO Box 2253
Los Banos, CA 93635

Account# 00-0025117
Account Manager: Ben Nydam
Submitted By: Lisa Machado

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:24

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1000-01	DW - MB	Well Water			12/13/2023 14:15
23L1000-02	DW - House	Well Water			12/13/2023 14:15

Default Cooler Temperature on Receipt °C: 0.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Received: 12/14/2023 14:20
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Sample Results

Sample: DW - MB
23L1000-01 (Water)

Sampled: 12/13/2023 14:15
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.42**	mmhos/cm	0.01	1		12/18/23 18:06	SM 2510 B		BEL0704
Electrical Conductivity umhos	**1420**	umhos/cm	10.0	1		12/18/23 18:06	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:15	Field		BEL0636
Nitrate Nitrogen as NO3N	**1.1**	mg/L	0.1	1	10	12/17/23 09:19	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 18:06	SM 4500-H+	H	BEL0704
pH	**8.0**	units	1.0	1		12/18/23 18:06	SM 4500-H+	H	BEL0704

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Sample Results
(Continued)

Sample: DW - House
23L1000-02 (Water)

Sampled: 12/13/2023 14:15
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.77**	mmhos/cm	0.01	1		12/18/23 18:08	SM 2510 B		BEL0704
Electrical Conductivity umhos	**773**	umhos/cm	10.0	1		12/18/23 18:08	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:15	Field		BEL0636
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	12/17/23 09:39	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 18:08	SM 4500-H+	H	BEL0704
pH	**7.8**	units	1.0	1		12/18/23 18:08	SM 4500-H+	H	BEL0704

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665										
Blank (BEL0665-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0665-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.2	90-110		
LCS (BEL0665-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
LCS (BEL0665-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.0	90-110		
LCS (BEL0665-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEL0665-DUP1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.639	10
Duplicate (BEL0665-DUP2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			0.787	10
Duplicate (BEL0665-DUP3)		**Source: 23L1000-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.612	10
Duplicate (BEL0665-DUP4)		**Source: 23L1005-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.7			0.941	10
Matrix Spike (BEL0665-MS1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L	5.000	5.8	102	90-110		
Matrix Spike (BEL0665-MS2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.3	102	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665 (Continued)										
Matrix Spike (BEL0665-MS3)			**Source: 23L1000-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.2	91.6	90-110		
Matrix Spike (BEL0665-MS4)			**Source: 23L1005-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.7	99.3	90-110		
Reference (BEL0665-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0665-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0665-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704										
Blank (BEL0704-BLK1)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0704-BLK2)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0704-BLK3)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Duplicate (BEL0704-DUP1)		**Source: 23L0999-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.01		0.01	mmhos/cm		1.01			0.248	10
pH	7.7		1.0	units		7.6			0.784	10
Electrical Conductivity umhos	1010		10.0	umhos/cm		1010			0.248	10
Duplicate (BEL0704-DUP2)		**Source: 23L1004-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.11		0.01	mmhos/cm		1.11			0.108	10
Electrical Conductivity umhos	1110		10.0	umhos/cm		1110			0.108	10
pH	7.8		1.0	units		7.7			0.775	10
Reference (BEL0704-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	426			umhos/cm	426.0		99.9	90-110		
Reference (BEL0704-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0704-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704 (Continued)										
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0704-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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12/14/23 14:20 23L1000

MMM

DELLAVALLE LABORATORY, INC.

WATER WORK REQUEST

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

Bill To: Acct No. 25117 Cons. 8

Purchase Order No. Results Needed By

No. of Samples 2 No. Bottles

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Client **JDL Dairy Inc**
Address 22495 W China Camp Road
City, State, Zip Los Banos, CA 93635
Phone Fax
Cell/Email

Copy to Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by Lisa Machado - (209) 704-5938

Ranch

Date sampled 12/13/23

Sampled by

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	2:15	Ø	0.7
2. DW-House	Sampled From:	12/13	2:15	Ø	2.9
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	mDekeyrel	CASinc	12/13/23 @ 2:15	12/14/23 12:4
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping
$ ___ In
$ ___ Out
Date

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

CC0E1E6

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1								
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1000

SEND OUT